Shoulder MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Core Measurements

System	Use
Ellman	PT cuff depth: 1 <25%, 2 = 25-50%, 3 >50%.
DeOrio	FT tear size: small <1 cm, medium 1-3 cm, large 3-5 cm, massive >5 cm.
Patte	Retraction: 1 lateral to head, 2 humeral dome, 3 glenoid/medial.
Goutallier	0 none, 1 streaks, 2 muscle>fat, 3 muscle=fat, 4 fat>muscle.
AHI	Normal 7-14 mm; cuff arthropathy often <6 mm.

Glenoid Track

D = expected inferior glenoid diameter; d = anterior glenoid bone loss. GT = 0.83D - d . HSI = Hill-Sachs width + intact medial bone bridge. Off-track if HSI >= GT.

How to Find It — Landmark-Based Identification

This section trains you to navigate a shoulder MRI by anchoring to fixed bony and soft-tissue landmarks, then working outward to each structure on the plane and sequence that shows it best. Master the three planes first, then use them to systematically interrogate the rotator cuff, biceps–labral complex, ligaments, and the neurovascular spaces.

The Three Standard Planes

📍 LANDMARK **Plane orientation**

Standard shoulder MRI uses three planes prescribed off the scout, all referenced to the scapular body and the **supraspinatus tendon**.

- **Oblique coronal**: prescribed parallel to the **supraspinatus muscle/tendon** (or perpendicular to the glenoid). Best for the supraspinatus, the superior labrum, the AC joint, and the supraspinatus footprint on the greater tuberosity.
- **Oblique sagittal**: perpendicular to the oblique coronal (parallel to the glenoid face). Best for the **rotator cuff in cross-section** (the "horseshoe" of cuff tendons), rotator interval, acromial morphology, and the spinoglenoid/suprascapular notches.

- **Axial**: best for the **subscapularis**, the **long head biceps in the bicipital groove**, the anterior and posterior labrum, and the glenohumeral ligaments.

💡 PEARL **Match the question to the plane**

Before calling anything, ask which plane answers your question. Supraspinatus footprint → coronal. Subscapularis and biceps groove → axial. Cuff muscle bulk/atrophy and interval → sagittal. Confirm every finding on at least two planes before you commit to it.

🔍 HOW TO FIND IT **ABER (abduction–external rotation) view**

If your protocol includes **ABER**, the arm is abducted with the hand behind the head. This tensions the **anterior band of the IGHL** and the anteroinferior labrum, and relaxes the cuff over the greater tuberosity.

- Best for: anteroinferior labral tears (Bankart family) and **articular-side partial supraspinatus tears (PASTA)** that gape open under tension
- Read it as an oblique-axial through the glenohumeral joint with the humeral head externally rotated

Rotator Cuff Tendons and Footprints

🔍 HOW TO FIND IT **Supraspinatus tendon**

On the **oblique coronal**, trace the muscle belly laterally over the humeral head to its footprint on the **superior facet of the greater tuberosity**.

- Normal: uniform low signal with a smooth taper to insertion
- Scrutinize the **critical zone** ~1 cm proximal to the footprint, the watershed where degenerative tears begin
- Confirm any signal abnormality on the **oblique sagittal**, where the supraspinatus sits at the "12 o'clock" position of the cuff horseshoe

⚠️ PITFALL **Magic-angle artifact in the supraspinatus**

On short-TE sequences (T1, PD, GRE), the obliquely oriented distal supraspinatus near its insertion can show artifactual intermediate signal where the fibers lie ~55° to B_0. This **disappears on T2** (long TE). Do not call a tear on a short-TE sequence alone.

🔍 HOW TO FIND IT **Infraspinatus tendon**

Posterior to the supraspinatus. On the **oblique sagittal**, it occupies the posterosuperior cuff just below supraspinatus; on **axial/coronal** images it inserts on the **middle facet of the greater tuberosity**, posterior to the supraspinatus footprint.

- Normal: low signal, broad tendon

- The supraspinatus and infraspinatus footprints overlap; many "supraspinatus" tears extend posteriorly into the infraspinatus — check the sagittal for the true anteroposterior extent

🔍 HOW TO FIND IT **Teres minor**

The most inferior posterior cuff muscle. On the **oblique sagittal**, find it at the bottom of the posterior horseshoe; it inserts on the **inferior facet of the greater tuberosity**.

- Normal: low-signal tendon, small muscle
- Its bulk and signal are the key clue to **axillary nerve / quadrilateral space** pathology (see below)

🔍 HOW TO FIND IT **Subscapularis tendon**

The anterior cuff. Best seen on **axial** images: trace the muscle from the anterior scapula laterally to its insertion on the **lesser tuberosity**.

- Normal: multipennate tendon with thin fat striations between fascicles (do not mistake the fat clefts for a split tear)
- Evaluate the upper fibers on the **sagittal**; tears typically begin at the superior insertion
- Confirm: a torn subscapularis often coexists with **biceps pulley failure and medial biceps subluxation**

📍 LANDMARK **Greater vs lesser tuberosity facets**

The **greater tuberosity** carries three facets — superior (supraspinatus), middle (infraspinatus), inferior (teres minor), front to back. The **lesser tuberosity** (anteromedial) is the subscapularis footprint. The **bicipital groove** runs between them and is your anchor for the biceps tendon.

📍 LANDMARK **Rotator cable and crescent**

The **rotator cable** is a thick band of coracohumeral ligament fibers arching across the undersurface of the cuff, perpendicular to the tendon fibers, seen best on **coronal/sagittal** as a low-signal cord medial to the footprint. The thinner **rotator crescent** lies lateral to it, between the cable and the insertion.

- Teaching point: an intact cable can keep the cuff functional ("suspension bridge" effect) even with a crescent tear — note cable integrity when describing tears

Long Head of the Biceps and the Biceps–Labral Complex

🔍 HOW TO FIND IT **Long head of biceps tendon (LHBT)**

Start on the **axial** at the **bicipital groove** between the tuberosities — the tendon is a round low-signal dot within a small amount of fluid. Follow it proximally as it turns into the joint, crosses under the **biceps pulley**, and runs to its anchor at the superior glenoid.

- Normal: uniform low signal, centered in the groove

- A little fluid in the sheath is normal (it communicates with the joint); fluid *only* around the biceps with a dry joint suggests tenosynovitis

🔍 HOW TO FIND IT **Biceps anchor / biceps–labral complex**

On the **oblique coronal**, follow the intra-articular biceps to where it blends with the **superior labrum at 12 o'clock**. This junction is the biceps–labral complex and the site of SLAP lesions.

⚠️ PITFALL **Empty groove sign**

An **empty bicipital groove** on axial means the LHBT is absent from its normal position — either a complete tear with retraction or **medial dislocation** out of the groove. Look medially and along the subscapularis for the displaced tendon before concluding.

Rotator Interval and the Biceps Pulley

📍 LANDMARK **Rotator interval**

The triangular gap between the **anterior supraspinatus** and the **superior subscapularis**, bounded medially by the coracoid. Read it on the **oblique sagittal** (most anterosuperior slices) and **axial**. It contains the **biceps pulley**: the **superior glenohumeral ligament (SGHL)** and the **coracohumeral ligament (CHL)**, which together sling the LHBT as it enters the groove.

🔍 HOW TO FIND IT **Biceps pulley (SGHL + CHL)**

On **axial** at the level of the proximal groove, the **SGHL** forms a thin band medial/deep to the biceps, and the **CHL** lies superficial. Together they form a sling stabilizing the biceps entrance.

- Evaluate pulley integrity whenever there is biceps instability or a high subscapularis tear

⚠️ PITFALL **Medial biceps subluxation and the comma sign**

When the pulley and superior subscapularis fail, the LHBT subluxates medially, often dragging torn subscapularis/CHL fibers with it. On axial, the displaced fibers curving off the lesser tuberosity form the **comma sign**, pointing you to the avulsed superomedial subscapularis. Always check the groove for biceps position when you see a subscapularis tear.

The Labrum — Clock-Face Method

💡 PEARL **Clock-face convention**

Use the face-on glenoid (oblique sagittal mindset, right shoulder): **12 = biceps anchor (superior)**, **3 = anterior**, **6 = inferior**, **9 = posterior**. State laterality, because 3 and 9 swap meaning between left and right shoulders. Localize every labral finding to a clock position and a plane.

🔍 HOW TO FIND IT **Reading the labrum across planes**

The labrum is normally low-signal and triangular, firmly attached to the glenoid rim.

- **Axial**: best for **anterior (3 o'clock)** and **posterior (9 o'clock)** labrum
- **Oblique coronal**: best for **superior (12)** and **inferior (6)** labrum and SLAP lesions
- On arthrography, contrast tracking *into* or *under* the labrum at an abnormal position/orientation indicates a tear

🔍 HOW TO FIND IT **SLAP lesion (superior labrum)**

On the **oblique coronal**, look for contrast/fluid signal extending into the **superior labrum at 12 o'clock**, laterally oriented and often extending posteriorly. SLAP tears propagate from the biceps anchor.

🔍 HOW TO FIND IT **Bankart-family lesions (anteroinferior labrum)**

Centered at the **3–6 o'clock anteroinferior** labrum, best on **axial** and **ABER**.

- **Bankart**: anteroinferior labral tear (± bony "bony Bankart" rim fracture)
- **Perthes**: labrum torn but periosteum intact, so the labrum is medially stripped but non-displaced — subtle; ABER helps lift it open
- **ALPSA**: anteroinferior labroligamentous sleeve avulsed and displaced medially, healing in a malpositioned location

⚠️ PITFALL **Normal variants — do NOT call these tears**

Several anterosuperior variants mimic tears. They all live **above the 3 o'clock equator** with smooth margins:

- **Sublabral foramen**: unattached anterosuperior labrum (1–3 o'clock) with intact rest of the labrum
- **Buford complex**: absent anterosuperior labrum + a cord-like thickened **MGHL** — never call the absent labrum a tear
- **Sublabral recess (sulcus)**: smooth, medially-oriented cleft under the **superior** labrum at 12, following the glenoid contour (vs a SLAP, which is irregular and laterally oriented and extends posteriorly)

Rule of thumb: a variant is smooth and parallels the glenoid; a tear is irregular and points laterally/away from the rim. **Below the equator (3–6–9), be very suspicious — true variants are uncommon there.**

Glenohumeral Ligaments

🔍 **HOW TO FIND IT** **Inferior glenohumeral ligament (IGHL) complex**

The primary anterior stabilizer in abduction. On **coronal/axial**, identify the **anterior band**, **posterior band**, and the dependent **axillary pouch** slung beneath the joint.

- Best tensioned and assessed on **ABER**
- A **HAGL** lesion is humeral-side avulsion of the IGHL — look for the axillary pouch contour changing from a normal **"U" into a "J"** as contrast leaks from the inferior capsule

🔍 **HOW TO FIND IT** **Middle glenohumeral ligament (MGHL)**

On **axial** anteriorly, a band running obliquely across the front of the joint, deep to the subscapularis. May be cord-like (normal variant, especially with a **Buford complex**).

🔍 **HOW TO FIND IT** **Superior glenohumeral ligament (SGHL)**

Small anterosuperior ligament forming part of the **biceps pulley** in the rotator interval; assess on **axial** at the level of the coracoid base/proximal biceps (see Rotator Interval).

AC Joint, Acromion, and Os Acromiale

🔍 **HOW TO FIND IT** **Acromioclavicular (AC) joint**

Best on **oblique coronal** (and sagittal). Assess for capsular hypertrophy, osteophytes projecting inferiorly, and marrow edema.

- Inferiorly-projecting AC osteophytes narrowing the supraspinatus outlet are a clue to **extrinsic impingement**

📍 **LANDMARK** **Acromial morphology (Bigliani)**

Judge acromial undersurface shape on the **oblique sagittal** outlet view: **Type I flat**, **Type II curved**, **Type III hooked** (most associated with outlet impingement). A **lateral downsloping** acromion also narrows the outlet.

⚠️ **PITFALL** **Os acromiale**

An unfused acromial ossification center. On **axial**, look for a corticated cleft across the **anterior acromion** — do not mistake it for a fracture. The mobile fragment can tilt down and contribute to impingement; note marrow edema across the synchondrosis if symptomatic.

Neurovascular Spaces and Denervation Patterns

🔍 HOW TO FIND IT **Suprascapular notch (suprascapular nerve)**

On the **oblique sagittal** medially, find the notch at the superior scapular margin. A cyst here (often from a labral tear) compresses the **suprascapular nerve** proximally, denervating **both supraspinatus and infraspinatus**.

🔍 HOW TO FIND IT **Spinoglenoid notch (suprascapular nerve, distal)**

On the **oblique sagittal/axial**, find the notch at the lateral base of the scapular spine. A **paralabral cyst** here (classically from a posterosuperior labral tear) compresses the nerve distally, denervating the **infraspinatus only** (supraspinatus spared).

- Teaching: isolated infraspinatus edema/atrophy → look for a spinoglenoid notch cyst and a posterior labral tear

🔍 HOW TO FIND IT **Quadrilateral space (axillary nerve)**

Bounded by **teres minor** (superior), **teres major** (inferior), **long head triceps** (medial), and the **humeral neck** (lateral); best appreciated on **sagittal/axial** posteriorly. The **axillary nerve** runs here.

- Denervation of **teres minor ± deltoid** (edema then fatty atrophy, especially isolated teres minor) points to **quadrilateral space syndrome**

💡 PEARL **Denervation localizes the lesion**

Match the denervated muscle to the nerve and the site: **supraspinatus + infraspinatus** → suprascapular nerve at the suprascapular notch; **infraspinatus alone** → suprascapular nerve at the spinoglenoid notch; **teres minor (± deltoid)** → axillary nerve in the quadrilateral space.

Bone Landmarks and Their Mimics

⚠️ PITFALL **Bare area of the humeral head**

On posterior coronal images, the **bare area** of the humeral head shows normal central cartilage thinning and small irregularities. Do not mistake it for a Hill-Sachs lesion or osteochondral defect — it is in the wrong location (central/posterosuperior bare area, not the posterolateral instability site).

📍 LANDMARK **Typical Hill-Sachs location**

A **Hill-Sachs** impaction is a wedge defect on the **posterosuperior–lateral humeral head**, seen on the **uppermost axial slices** (at or above the coracoid). It pairs with anterior instability and a Bankart lesion.

- Rule of thumb: a humeral head defect **at the level of the coracoid or above** on axial is in the Hill-Sachs zone; defects below that level are usually the **normal bare area** and should not be called.

⚠ PITFALL **Glenoid bare area / normal rim irregularity**

The **central glenoid** can show normal cartilage thinning (a "bare spot"). Keep labral and bony Bankart assessment at the **rim**, not the central glenoid face, to avoid over-calling cartilage loss as pathology.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace [surface/compartment] with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

F: Articular cartilage of the [surface/compartment] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

I: No chondral defect of the [surface/compartment].

Grade 1 - chondral signal change / softening

F: Mild focal cartilage signal heterogeneity/softening of the [surface/compartment] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

I: Grade 1 chondromalacia of the [surface/compartment].

Grade 2 - superficial partial-thickness chondral loss

F: Superficial chondral fissuring/fraying of the [surface/compartment] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

I: Grade 2 chondromalacia of the [surface/compartment].

Grade 3 - deep partial-thickness chondral loss

F: Deep partial-thickness chondral fissuring/defect of the [surface/compartment] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

I: Grade 3 chondromalacia of the [surface/compartment].

Grade 4 - full-thickness chondral loss

F: Full-thickness cartilage loss/fissuring of the [surface/compartment] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change]. [Small unstable chondral flap/loose body if present.]

I: Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartment].

Rotator Cuff Tells

- Tendinosis: thickened/intermediate signal, no fluid cleft.
- Tear: fluid-equivalent cleft or tendon discontinuity.
- Subcortical GT cyst: search adjacent articular-sided cuff tear.
- Delamination: longitudinal fluid cleft paralleling fibers.
- Cable intact in large tear: potentially compensated mechanics.
- Tangent sign: supraspinatus belly below coracoid-spine line = atrophy.

Subscap/Biceps Pulley

- LHBT centered in groove: superior subscap/pulley usually intact.
- Medial LHBT subluxation: superior subscap + pulley injury until proven otherwise.
- Comma sign: retracted SGHL/CHL/superior subscap band medial to joint/coracoid.
- Lesser tuberosity cyst/uncovering: indirect subscap tear sign.

Instability Tells

- Bankart: anteroinferior labroligamentous detachment.
- Perthes: torn labrum but periosteum intact; may look nondisplaced.
- ALPSA: labrum displaced medially/inferiorly along glenoid neck.
- GLAD: superficial labral tear + adjacent glenoid cartilage defect; pain more than instability.
- HAGL: IGHL avulsed from humerus, J-shaped axillary pouch, medial humeral neck fluid/contrast.
- Internal impingement: articular posterior cuff tear + posterosuperior labral tear in overhead athlete.

Normal / Negative Templates

Normal rotator cuff

F: Rotator cuff tendons are intact without tendinosis, partial-thickness tear, or full-thickness tear. Muscle bulk and signal are preserved without fatty atrophy.

I: Intact rotator cuff. No muscle atrophy.

Negative labrum on nonarthrogram

F: No displaced labral tear or paralabral cyst identified on this nonarthrographic examination.

I: No discrete labral tear identified by nonarthrographic MRI.

Useful Caveats

Labrum: If no joint distention, avoid overcalling subtle labral signal unless displaced tear/cyst/chondral injury.

Post-op cuff: Tendon thickening and heterogeneous signal can persist after repair; diagnose retear when there is fluid signal traversing tendon to both articular and bursal surfaces or clear tendon gap.

Subcoracoid: Narrow coracohumeral interval alone is weak; report compression/tendinosis/tear of subscapularis and clinical correlation.

ROTATOR CUFF - TENDINOSIS, PARTIAL TEARS, FULL-THICKNESS TEARS

Tendinopathy / Calcific

Mild tendinosis

F: Mild intermediate intrasubstance signal of the distal [supraspinatus] tendon without discrete fluid-equivalent cleft. Tendon thickness is preserved.

I: Mild [supraspinatus] tendinosis. No tear.

Moderate tendinosis

F: Heterogeneous intermediate-to-mildly hyperintense signal and mild thickening of the distal [tendon]. No fluid-signal tendon defect.

I: Moderate [tendon] tendinosis without tear.

Severe tendinosis

F: Marked globular thickening and diffuse intermediate intrasubstance signal of the distal [tendon] with loss of normal fibrillar architecture. No definite fluid-equivalent cleft.

I: Severe [tendon] tendinosis. Superimposed low-grade interstitial tearing is difficult to exclude.

Calcific tendinitis

F: Globular T1/T2 hypointense focus within the distal [supraspinatus] tendon measuring [X] mm with blooming susceptibility. Adjacent reactive tendinopathy and [mild] subacromial-subdeltoid bursal edema.

I: Calcific tendinitis of the [supraspinatus] with reactive [subacromial-subdeltoid bursitis].

Partial Thickness

Low-grade articular-sided tear

F: Focal fluid-signal cleft at the articular surface of the distal [supraspinatus] footprint involving less than 25% of tendon thickness. Bursal surface fibers remain intact.

I: Low-grade (<25%) partial-thickness articular surface tear of the [supraspinatus].

Intermediate articular-sided tear

F: Articular-surface fluid cleft at the [supraspinatus] footprint measuring [X] mm AP by [Y] mm mediolateral and involving 25-50% of tendon thickness. Bursal surface remains intact.

I: Intermediate-grade (25-50%) partial-thickness articular surface tear of the [supraspinatus].

High-grade PASTA

F: High-grade articular-sided fluid defect at the [supraspinatus] footprint extending greater than 50% through tendon thickness with adjacent greater tuberosity cortical irregularity/cystic change. Thin bursal fibers remain intact.

I: High-grade (>50%) partial articular surface tendon avulsion (PASTA) of the [supraspinatus], near full-thickness.

Bursal-sided partial tear

F: Bursal-sided fluid-signal cleft of the distal [supraspinatus] tendon involving approximately [X]% of tendon thickness with adjacent subacromial-subdeltoid bursal fluid.

I: [Low/intermediate/high]-grade partial-thickness bursal surface tear of the [supraspinatus] with reactive bursitis.

Interstitial / delaminating tear

F: Longitudinal fluid-signal cleft within the substance of the distal [tendon], paralleling the tendon fibers and not clearly extending to either articular or bursal surface.

I: Intrasubstance delaminating tear of the [tendon].

Full-Thickness Tears

Small full-thickness tear

F: Full-thickness fluid-signal defect of the distal [supraspinatus] footprint measuring [X] mm AP by [Y] mm mediolateral. Minimal medial tendon retraction. Fluid communicates with the subacromial-subdeltoid bursa.

I: Small (<1 cm) full-thickness [supraspinatus] tear without significant retraction.

Medium full-thickness tear

F: Full-thickness [supraspinatus] tear measuring [X] cm AP by [Y] cm mediolateral with tendon stump retracted to the humeral head dome (Patte 2). Muscle bulk is preserved with Goutallier [0-1] fatty infiltration.

I: Medium (1-3 cm) full-thickness [supraspinatus] tear with mild retraction and no significant muscle atrophy.

Massive chronic tear

F: Full-thickness, full-width tear of the [supraspinatus and infraspinatus] measuring [X] cm with tendon stump retracted to the glenoid/medial glenoid (Patte 3). Moderate-to-severe muscle volume loss with Goutallier [3/4] fatty infiltration. Acromiohumeral interval measures [X] mm.

I: Chronic massive full-thickness [supraspinatus/infraspinatus] tear with advanced retraction and fatty atrophy, likely irreparable.

Cuff Arthropathy

Cuff tear arthropathy

F: Chronic massive rotator cuff tear with superior humeral head migration, acromiohumeral interval [X] mm, acetabularization of the undersurface of the acromion, femoralization of the humeral head, and glenohumeral chondral loss/subchondral cystic change.

I: Rotator cuff tear arthropathy, Hamada grade [3/4/5].

Subacromial-subdeltoid bursitis

F: Small/moderate fluid and synovial thickening in the subacromial-subdeltoid bursa. [No full-thickness cuff tear identified.]

I: Subacromial-subdeltoid bursitis [with/without rotator cuff tear].

SUBSCAPULARIS, BICEPS PULLEY, AND LONG HEAD BICEPS

Subscapularis Tears

Superior-facet low-grade subscap tear

F: Focal fluid-signal cleft at the superolateral subscapularis insertion on the lesser tuberosity involving the cephalad fibers/first facet. Remaining inferior subscapularis fibers are intact. Long head biceps tendon remains normally seated in the bicipital groove.

I: Low-grade partial-thickness tear of the cephalad subscapularis tendon. Biceps pulley remains intact.

High-grade cephalad subscap + medial LHBT subluxation

F: High-grade tear of the cephalad subscapularis tendon with fluid gap/uncovering of the superior lesser tuberosity. The long head biceps tendon is medially subluxed from the bicipital groove, compatible with biceps reflection pulley disruption.

I: High-grade cephalad subscapularis tear with biceps pulley disruption and medial subluxation of the long head biceps tendon.

Full-thickness retracted subscap + comma sign

> **F:** Full-thickness tear of the superior [and mid] subscapularis tendon with medial/superomedial tendon retraction. A curvilinear low-signal soft tissue band in the rotator interval medial to the joint/coracoid is compatible with a comma sign, representing retracted superior subscapularis fibers and torn SGHL/CHL. The long head biceps tendon is medially dislocated with an empty bicipital groove.

> **I:** Full-thickness retracted subscapularis tendon tear with comma sign, biceps pulley disruption, and medial dislocation of the long head biceps tendon.

Chronic subscap tear

> **F:** Chronic full-thickness subscapularis tendon tear with medial retraction, lesser tuberosity uncovering/cystic change, and [moderate/severe] subscapularis muscle atrophy/fatty infiltration.

> **I:** Chronic full-thickness subscapularis tear with advanced fatty atrophy.

Pulley / LHBT

Pulley sprain without biceps instability

> **F:** Edema and irregularity of the rotator interval/biceps reflection pulley without medial subluxation of the long head biceps tendon. Cephalad subscapularis fibers are intact.

> **I:** Low-grade biceps pulley sprain without biceps instability.

LHBT tendinosis

> **F:** Long head biceps tendon is thickened with intermediate intrasubstance signal in the groove/intra-articular segment. No tendon discontinuity or subluxation.

> **I:** Long head biceps tendinosis. No tear or instability.

LHBT tenosynovitis

> **F:** Disproportionate fluid surrounding an otherwise intact long head biceps tendon within the bicipital groove. Tendon caliber and signal are preserved.

> **I:** Long head biceps tenosynovitis.

LHBT split tear

> **F:** Long head biceps tendon is thickened and irregular with longitudinal intrasubstance fluid-signal cleft/splitting in the bicipital groove. Tendon remains continuous.

> **I:** Longitudinal split tear/high-grade partial tear of the long head biceps tendon.

Complete proximal LHBT tear

> **F:** Empty bicipital groove with nonvisualization of the intra-articular long head biceps tendon. Distally retracted tendon stump is present at the [proximal humerus/mid arm].

> **I:** Complete proximal long head biceps tendon tear with distal retraction.

Subcoracoid / Anterosuperior Impingement

Subcoracoid impingement pattern

F: Narrowing of the coracohumeral interval to [X] mm with crowding/effacement of the subcoracoid fat, subscapularis tendinosis, and articular-sided partial tearing of the cephalad subscapularis tendon. [Reactive subcoracoid bursal fluid.]

I: Findings support subcoracoid impingement with cephalad subscapularis tendinosis/partial tearing. Correlate with anterior shoulder pain and provocative exam.

Anterosuperior internal impingement

F: Articular-sided fraying/partial tearing of the superior subscapularis tendon and biceps pulley complex with adjacent anterosuperior labral degeneration. Long head biceps tendon is [intact/subluxed/tendinotic].

I: Anterosuperior internal impingement pattern involving the superior subscapularis/pulley complex.

LABRUM, INSTABILITY, HAGL, GLAD, AND GLENOID TRACK

Anterior Instability

Soft tissue Bankart

F: Detachment of the anteroinferior labrum and attached inferior glenohumeral ligament from the glenoid rim at [3-6 o'clock]. No displaced osseous fragment. [Associated Hill-Sachs impaction fracture.]

I: Soft tissue Bankart lesion of the anteroinferior labrum [with Hill-Sachs lesion], compatible with prior anterior shoulder dislocation.

Bony Bankart

F: Anteroinferior glenoid rim fracture/labral avulsion involving approximately [X]% of glenoid width/surface area. Fragment is [displaced/nondisplaced].

I: Bony Bankart lesion with approximately [X]% anterior glenoid bone loss.

ALPSA

F: Anteroinferior labroligamentous complex is stripped and displaced medially/inferiorly along the glenoid neck with intact scapular periosteal sleeve.

I: ALPSA lesion of the anteroinferior labroligamentous complex.

Perthes

F: Nondisplaced tear of the anteroinferior labrum with intact but stripped scapular periosteum. Labrum remains near the glenoid rim.

I: Perthes variant anteroinferior labral tear.

HAGL / IGHL

HAGL lesion

> **F:** Inferior glenohumeral ligament is avulsed from its humeral attachment with lax, edematous, J-shaped configuration of the axillary pouch. Fluid/contrast extravasates along the medial humeral neck. Anteroinferior labrum is [intact/torn].

> **I:** Humeral avulsion of the inferior glenohumeral ligament (HAGL lesion), a cause of anterior glenohumeral instability.

BHAGL lesion

> **F:** Avulsion of the inferior glenohumeral ligament from the humeral attachment with small osseous avulsion fragment from the medial humeral neck and surrounding edema/fluid extravasation.

> **I:** Bony humeral avulsion of the inferior glenohumeral ligament (BHAGL lesion).

Posterior HAGL

> **F:** Posterior band of the inferior glenohumeral ligament is avulsed from its humeral attachment with posterior capsular discontinuity and fluid/contrast tracking along the posteromedial humeral neck.

> **I:** Posterior humeral avulsion of the inferior glenohumeral ligament (posterior HAGL lesion).

Search pattern: HAGL can occur with little/no labral tear. Check subscapularis and LHBT because associated cuff tears commonly involve subscapularis.

GLAD / Chondrolabral

Anterior GLAD

> **F:** Superficial nondisplaced tear of the anteroinferior labrum with adjacent focal cartilage defect/fissuring of the anteroinferior glenoid. Deep labral attachment is relatively preserved. No displaced Bankart fragment.

> **I:** Glenolabral articular disruption (GLAD lesion) of the anteroinferior glenoid/labrum.

Posterior GLAD

> **F:** Superficial tear of the posterior/posteroinferior labrum with adjacent focal posterior glenoid chondral defect and mild subchondral marrow edema/cystic change.

> **I:** Posterior GLAD lesion involving the posterior labrum and adjacent glenoid articular cartilage.

Glenoid chondral flap

> **F:** Focal delaminating chondral flap at the [anteroinferior/posterior/superior] glenoid measuring [X] mm with adjacent labral fraying/tear and trace subchondral marrow edema.

> **I:** Focal glenoid chondral flap/delamination with adjacent [labral tear].

SLAP / Posterior Labrum / Paralabral Cyst

SLAP type II

F: Irregular laterally curved fluid-signal cleft undermines the superior labrum and extends posterior to the biceps anchor with detachment of the biceps-labral anchor from the supraglenoid tubercle.

I: Type II SLAP tear with biceps-labral anchor detachment.

Posterior labral tear / reverse Bankart

F: Detachment of the posteroinferior labrum from the glenoid rim with [posterior capsular stripping/reverse bony Bankart fragment]. Mild posterior humeral head subluxation.

I: Posteroinferior labral tear/reverse Bankart lesion with posterior instability pattern.

Paralabral cyst with nerve effect

F: Multiloculated paralabral cyst at the [spinoglenoid/suprascapular] notch measuring [X] cm, contiguous with a [posterior/superior] labral tear. There is denervation edema/atrophy of the [infraspinatus only/supraspinatus and infraspinatus].

I: [Posterior/superior] labral tear with paralabral cyst causing [spinoglenoid/suprascapular] notch neuropathy.

Hill-Sachs and Glenoid Track

On-track Hill-Sachs

F: Posterosuperolateral humeral head impaction fracture. Glenoid track measures [GT] mm; Hill-Sachs interval measures [HSI] mm, which is less than the glenoid track.

I: On-track Hill-Sachs lesion.

Off-track Hill-Sachs

F: Posterosuperolateral humeral head impaction fracture. Anterior glenoid bone loss measures [d] mm; expected glenoid diameter [D] mm; calculated glenoid track [GT] mm. Hill-Sachs interval [HSI] mm equals/exceeds the glenoid track.

I: Off-track Hill-Sachs lesion, at risk for engagement; consider bone augmentation/stabilization planning.

IMPINGEMENT PATTERNS, AC JOINT, ADHESIVE CAPSULITIS, ARTHRITIS

Subacromial / External Impingement

Outlet impingement pattern

F: [Type II/III hooked/laterally downsloping] acromion with undersurface acromial spur and mild acromioclavicular osteoarthrosis narrowing the supraspinatus outlet. There is subacromial-subdeltoid bursitis and bursal-sided fraying/partial tearing of the distal supraspinatus tendon.

I: Subacromial outlet impingement pattern with subacromial-subdeltoid bursitis and bursal-sided supraspinatus fraying/partial tear.

Impingement without tear

F: Mild subacromial-subdeltoid bursal fluid and supraspinatus tendinosis. No fluid-signal partial- or full-thickness rotator cuff tear.

I: Mild subacromial-subdeltoid bursitis/rotator cuff tendinosis. No cuff tear.

AC joint narrowing outlet

F: Moderate/severe acromioclavicular osteoarthrosis with inferior capsular hypertrophy/osteophytes contacting the myotendinous junction of the supraspinatus. Mild subacromial-subdeltoid bursal edema.

I: Acromioclavicular osteoarthrosis with inferior osteophytes contributing to supraspinatus outlet narrowing.

Internal Posterosuperior Impingement

Classic overhead athlete pattern

F: Articular-sided partial-thickness tearing/fraying of the posterior supraspinatus/anterior infraspinatus fibers at the footprint with adjacent posterosuperior labral fraying/tear. Mild cystic/edematous change is present in the posterosuperior humeral head/greater tuberosity. [Posterior capsular thickening.]

I: Findings consistent with internal posterosuperior impingement, including articular-sided posterior cuff tear and posterosuperior labral pathology.

Internal impingement with Bennett lesion

F: Posterior capsular thickening and chronic ossification/mineralization along the posteroinferior glenoid rim compatible with Bennett lesion. Associated posterosuperior labral tear and articular-sided posterior cuff fraying.

I: Chronic internal impingement/posterior capsular traction pattern with Bennett lesion, posterosuperior labral tear, and articular-sided posterior cuff fraying.

Adhesive Capsulitis / Capsule

Adhesive capsulitis

F: Thickening and edema of the axillary pouch/inferior glenohumeral ligament measuring [X] mm with edema/scarring in the rotator interval and partial effacement of the subcoracoid fat triangle.

I: MRI findings compatible with adhesive capsulitis.

Capsular sprain

F: Edema and thickening of the [anterior/inferior/posterior] capsule without focal capsular discontinuity or contrast extravasation.

I: [Anterior/inferior/posterior] capsular sprain without discrete capsular tear.

Glenohumeral / AC Arthritis

Glenohumeral OA

F: Diffuse glenohumeral chondral thinning with marginal osteophytes, subchondral marrow edema/cystic change, and degenerative labral tearing. Small joint effusion with synovitis.

I: [Mild/moderate/severe] glenohumeral osteoarthrosis with degenerative labral tearing and synovitis.

AC arthrosis active

F: Acromioclavicular joint space narrowing with capsular hypertrophy, marginal osteophytes, and prominent subchondral marrow edema across the joint.

I: Active acromioclavicular osteoarthrosis with periarticular marrow edema.

Distal clavicle osteolysis

F: Distal clavicular subchondral marrow edema, cortical irregularity/resorption, and AC joint capsular hypertrophy without acute fracture.

I: Distal clavicle osteolysis, typically repetitive stress related.

POSTOPERATIVE SHOULDER MRI - CUFF REPAIR, BICEPS TENODESIS, STABILIZATION

Postoperative Rotator Cuff

Intact cuff repair

F: Postsurgical changes of [supraspinatus] tendon repair with suture anchors in the greater tuberosity. Repaired tendon is continuous to the footprint with expected postoperative thickening/heterogeneous signal. No fluid-signal gap traversing the tendon. No anchor displacement.

I: Intact [supraspinatus] repair. Expected postoperative tendon heterogeneity without retear.

Low-grade postoperative interstitial signal

> **F:** Repaired [supraspinatus] tendon is continuous with mild heterogeneous/interstitial increased signal near the footprint, favored postoperative granulation/scar and tendinosis. No full-thickness fluid gap or tendon retraction.

> **I:** Intact cuff repair with postoperative tendinosis/scarring. No full-thickness retear.

Partial-thickness retear

> **F:** Postsurgical cuff repair with focal articular/bursal-sided fluid-signal defect at the repaired [supraspinatus] footprint involving approximately [X]% of tendon thickness. No full-thickness gap or medial retraction.

> **I:** Partial-thickness retear of the repaired [supraspinatus] tendon.

Full-thickness retear

> **F:** Recurrent full-thickness fluid-signal defect through the repaired [supraspinatus] tendon measuring [X] cm AP by [Y] cm mediolateral with medial tendon retraction to [location]. Fluid communicates with the subacromial-subdeltoid bursa. Suture anchors are [intact/displaced].

> **I:** Full-thickness retear of the repaired [supraspinatus] tendon with [mild/moderate/severe] retraction.

Anchor complication

> **F:** Suture anchor at the [greater/lesser] tuberosity is proud/displaced with surrounding marrow edema/cystic change and adjacent bursal/synovial reaction. Repaired tendon is [intact/torn].

> **I:** Suspected suture anchor loosening/displacement with reactive marrow and synovial/bursal inflammation.

Biceps / Labral Stabilization

Biceps tenodesis intact

> **F:** Postsurgical long head biceps tenodesis at the [proximal humerus/bicipital groove/subpectoral humerus]. Tenodesis tendon is intact without hardware displacement or surrounding fluid collection.

> **I:** Intact biceps tenodesis.

Biceps tenodesis failure

> **F:** Nonvisualization/discontinuity of the tenodesed biceps tendon at the fixation site with distal tendon retraction and fluid/edema along the tenodesis tract. Hardware is [intact/displaced].

> **I:** Failed biceps tenodesis with tendon retraction.

Post Bankart repair intact

> **F:** Postsurgical changes of anterior labral repair with anchors at the anteroinferior glenoid. Labroligamentous complex is apposed to the glenoid rim without displaced recurrent tear or paralabral cyst. No progressive glenoid bone loss.

> **I:** Intact anterior labral stabilization repair. No displaced recurrent labral tear.

Recurrent labral tear after repair

F: Postsurgical anterior labral repair with recurrent fluid-signal cleft undermining the [anteroinferior] labrum and extending to the anchor margin. Labrum is [displaced/nondisplaced]. [Associated paralabral cyst/recurrent Hill-Sachs.]

I: Recurrent [anteroinferior] labral tear at the prior repair site.

Postoperative infection concern

F: Large joint effusion with diffuse synovial thickening, periarticular soft tissue edema, and marrow edema around [anchor/hardware]. [Rim-enhancing fluid collection if contrast.]

I: Marked postoperative synovitis/effusion; infection should be excluded clinically.

Search Pattern

- **1. Check history, indication, and priors.**
- **2. Check adequacy, technique, and limitations.**
- **3. Look at localizers** for incidental pathology (mass lesions, serious pathology easily missed).
- **4. Assess bone cortex and marrow:**
- Cortical breaks/fractures, Hill-Sachs/trough sign, osteophytes, loose bodies, heterotopic ossification.
- Marrow replacement (darker than muscle on T1 = infection/neoplasia concern).
- Hematopoietic marrow distribution.
- Incidental ribs/vertebra.
- **5. Assess fluid** in subacromial-subdeltoid bursa and glenohumeral joint; if arthrography, check for abnormal communication.
- **6. Assess AC joint:** osteophytes, os acromiale on axials, subacromial spur.
- **7. Assess tendons:**
- Start with supraspinatus (coronals + sagittal fluid-sensitive); look for fluid signal traversing tendon (full or partial tear — bursal, articular, interstitial).
- Correlate with SA-SD bursa fluid if complete tear and note retraction.
- Intermediate signal + thickening = tendinosis; hydroxyapatite = low signal.
- Then infraspinatus, teres minor, subscapularis.
- Biceps tendon: check bicipital groove position/size (subluxation implies subscapularis tear), intra-articular course.
- **8. Assess glenohumeral joint:**
- Alignment (superior displacement = chronic cuff tear, inferior = effusion/atrophy); effusion.
- Inferior glenohumeral ligament (thickening/signal on coronals = synovitis; loss of fat signal at rotator interval = adhesive capsulitis).
- Labrum and MGHL (sagittals as map, coronals and axials to assess; superior labrum for SLAP; anterosuperior labrum and Buford complex/sublabral recess; anteroinferior labrum for Bankart spectrum — any detachment of anteroinferior labrum is abnormal; posterior labrum; reciprocating lesions across joint).
- Cartilage (axial and coronal PD/fluid-sensitive; glenoid and humerus separately; subchondral marrow changes).

- **9. Assess coracoacromial and coracoclavicular ligaments:** sagittal images, discontinuity = AC joint injury.
- **10. Assess subcutaneous tissues and musculature:**
- T1 precontrast images; atrophy, edema, inflammation, mass.
- Rotator cuff atrophy (supraspinatus below level of spine on oblique sagittals = atrophy).
- **11. Check neurovascular structures:** brachial plexus, suprascapular notch, spinoglenoid notch, quadrilateral space (especially if denervation present); incidental lung/chest viscera.
- **12. Double check** marrow signal, muscles, and subcutaneous tissues.
- **13. Proofread report.**

Infection Search Pattern

- **1. Marrow/bone changes:**
- Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).
- Marrow edema (isolated finding raises concern in diabetics/high-risk).
- Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
- Rim-enhancing collection = abscess.
- Superficial lesion without enhancement = ulcer.
- Enhancement along fascial planes = infectious/inflammatory fasciitis.
- Non-enhancement of normal structures = ischemia.
- Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
- Joint effusion with marrow replacement on both sides of joint.
- Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
- Sinus tract connecting abnormal marrow signal to skin surface.
- Disappearance of subchondral cysts on sequential imaging.
- Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Source Notes

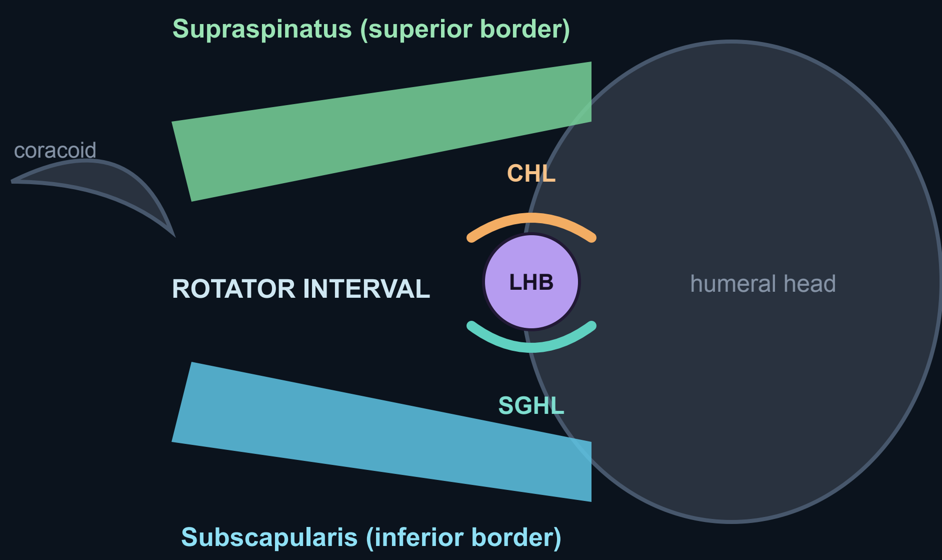
Built from the original one-page shoulder MRI cheat sheet plus Radsource Subscapularis Tendon Tear, Radsource HAGL Lesion, and review-level summaries of internal/subacromial/subcoracoid impingement MRI findings. Radsource subscapularis pearls used here: superolateral lesser tuberosity/first facet origin of many tears, lesser tuberosity uncovering/cysts, medial LHBT instability as an indirect sign, and comma sign composition from retracted superior subscapularis + SGHL/CHL. HAGL pearls: abnormal/discontinuous IGHL, J-shaped axillary pouch, medial humeral neck fluid/contrast, BHAGL fragment, and associated subscapularis/LHBT injuries.

Test Yourself & Visual Pearls

Lock in the high-yield shoulder anatomy and pathology with annotated diagrams and active-recall questions.

The rotator interval

The rotator interval is the triangular gap between the anterior margin of supraspinatus and the superior margin of subscapularis, through which the long head of biceps enters the joint.



Rotator interval. The triangular gap between the **supraspinatus** (superior) and **subscapularis** (inferior); the **CHL + SGHL** form the pulley sling that stabilizes the long head of biceps (LHB) entering the groove.

🔑 KEY POINTS **Rotator interval**

- **Contents:** SGHL, coracohumeral ligament (CHL), and the biceps pulley (SGHL/CHL sling stabilizing the LHB)
- **Borders:** supraspinatus superiorly, subscapularis inferiorly, coracoid medially at the apex
- **Function:** stabilizes the long head of biceps and restrains inferior/posterior translation
- **Pathology:** widening/scarring seen in adhesive capsulitis; pulley lesions destabilize the LHB tendon

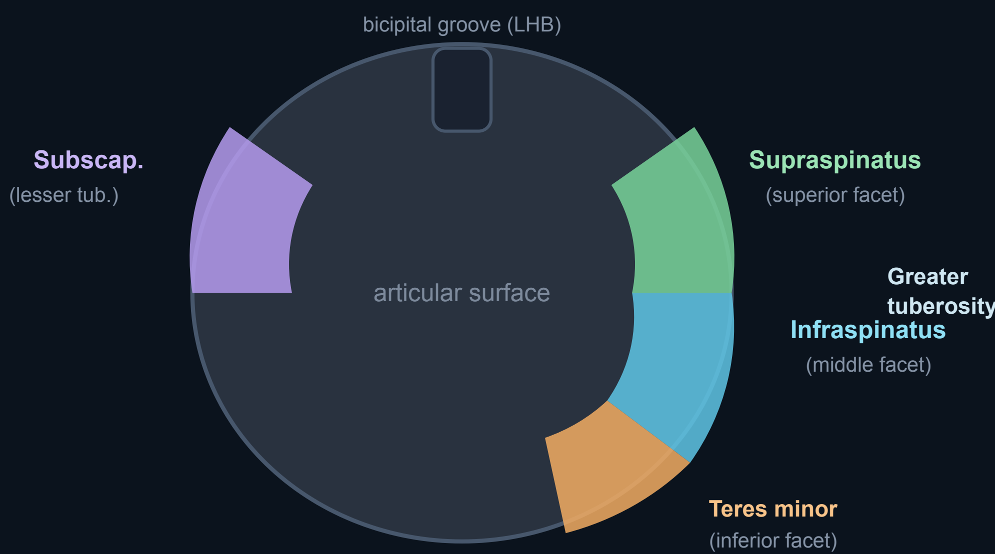
❓ QUIZ **Which two ligaments form the biceps pulley (reflection sling) within the rotator interval?**

❓ QUIZ **A tear of which cuff tendon is most commonly associated with medial dislocation of the biceps tendon out of the groove?**

Cuff footprints

Knowing the normal tendon insertions lets you localize a tear precisely and distinguish a true defect from anatomic variation.

PROXIMAL HUMERUS (top-down)



Rotator cuff footprints. Greater tuberosity facets — **supraspinatus** (superior), **infraspinatus** (middle), **teres minor** (inferior); **subscapularis** on the lesser tuberosity, with the bicipital groove between.

💡 TIP **Footprint anatomy**

Supraspinatus and infraspinatus insert on the greater tuberosity (superior and middle facets, with considerable overlap of the SS/IS footprint), teres minor on the inferior facet, and subscapularis on the lesser tuberosity. The bare area of the humeral head lies just posterior to the cuff insertion and should not be mistaken for an erosion.

❓ QUIZ **On which tuberosity and facet does the supraspinatus insert, and where does subscapularis go?**

Rapid review

Spaced-recall questions across the highest-yield shoulder topics — answer each before revealing.

? QUIZ **What is a PASTA lesion, and which surface of the cuff does it involve?**

? QUIZ **In a full-thickness supraspinatus tear, how do you grade retraction and assess reparability?**

? QUIZ **How is the tangent sign performed and what does a positive result mean?**

⚠ PITFALL **Goutallier fatty infiltration is not reversible**

Grade 3–4 fatty infiltration (fat equal to or greater than muscle) and a **positive tangent sign** predict a high rate of repair failure and irreparability. Unlike acute tears, advanced fatty muscle degeneration does **not** reverse after repair — flag it explicitly, as it changes surgical decision-making toward debridement, tendon transfer, or arthroplasty.

? QUIZ **Define the impingement terminology: subacromial (external) vs internal impingement.**

? QUIZ **How do you distinguish a SLAP tear from a normal sublabral recess?**

? QUIZ **Name the members of the "Bankart family" of anteroinferior labroligamentous lesions.**

⚠ PITFALL **Don't miss a HAGL or the J-sign**

A HAGL lesion (Humeral Avulsion of the Glenohumeral Ligament) avulses the IGHL from its **humeral** rather than glenoid side and is a frequently overlooked cause of recurrent instability. On coronal images the normal U-shaped axillary pouch becomes a J-shaped pouch (the "J-sign") as contrast/fluid leaks inferomedially. Always inspect the humeral attachment of the IGHL, not just the labrum.

? QUIZ **What makes a Hill-Sachs lesion "engaging," and what is the glenoid track concept?**

? QUIZ What are the phases of calcific tendinitis (hydroxyapatite deposition)?

? QUIZ What are the MRI hallmarks of adhesive capsulitis (frozen shoulder)?

? QUIZ How do you localize suprascapular nerve denervation: suprascapular notch vs spinoglenoid notch?